

Curated Research Articles

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- **Si@MXene hybrid anodes for lithium-ion batteries: Interface engineering, structural design, and mechanistic insights toward high-performance energy storage** — score: 1.000 This article discusses the development of silicon@MXene hybrid anodes for lithium-ion batteries, focusing on interface engineering and structural design to enhance performance. The authors provide mechanistic insights that contribute to improved energy storage capacity and efficiency in these advanced battery materials, positioning them as a potential solution for high-performance energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Operando X-ray Photoelectron Spectroscopy for Solid-State Batteries: Current Development and Future Perspectives** — score: 1.000 The article discusses the advancements in operando X-ray Photoelectron Spectroscopy (XPS) techniques specifically tailored for studying solid-state batteries. It highlights recent developments, potential applications, and future directions for this method in enhancing the understanding of battery materials and processes. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **In Situ Evolution of Artificial Interphase Enabling Stable, High-Performance Lithium Metal Batteries** — score: 1.000 This study introduces a dynamic artificial interphase for lithium metal batteries, created from a Mg₃N₂-based layer that evolves into a hybrid structure of Li₃N and Mg-Li alloy during cycling. This self-reinforcing mechanism significantly enhances battery stability by suppressing dendrite growth and enabling long-term cycling performance, achieving over 91% capacity retention after 600 cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Fluorine-Aligned Electrostatic Gatekeepers Regulating Polysulfide Migration for Stable Lithium-Sulfur Batteries** — score: 1.000 The article presents an innovative MXene-based separator that utilizes highly oriented C-F dipoles from -PVdF to create an electrostatic field that repels anionic polysulfides in lithium-sulfur batteries. This “push-back” mechanism effectively confines polysulfides to the cathode, enhancing cycling stability and performance while addressing limitations of conventional adsorption-based methods, thereby offering a new pathway for improving energy storage system durability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Impacts of the Particle Geometric Heterogeneity on Battery Degradation** — score: 1.000 This study introduces a novel AI visual imaging approach to investigate battery aging mechanisms by revealing hidden micro-scale interactions within particles. By employing deep learning to analyze the effects of geometric heterogeneity, the researchers demonstrate how microregion variations impact lithium-ion transport and cause battery degradation. The method not only deciphers complex aging processes but also enhances battery lifespan by optimizing electrode architecture, potentially informing design improvements in other energy systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Rapid Large-Scale Morphological Control and Interface Functionalization Enable All-Weather LiFePO₄ Cathode Material** — score: 0.900 The article introduces a scalable technique combining a super-liquid-film-reactor (SLFR) and nitrogen-containing coatings to enhance the performance of LiFePO₄ cathodes for lithium-ion batteries. This innovative approach significantly improves ionic diffusion and charge transport efficiency, particularly in cold conditions, resulting in high capacity retention and robust performance across a wide temperature range, demonstrating potential for industrial application. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Resolving the Kinetics-Stability Trade-Off in Prussian Blue Analogues for Aqueous Ca-Ion Batteries: A High-Entropy Vacancy Caging Strategy** — score: 0.800 The article discusses a novel approach using high-entropy vacancy caging to address the trade-off between kinetics and stability in Prussian blue analogues, enhancing their performance as cathode materials for aqueous calcium-ion batteries. This strategy aims to optimize the efficiency of these batteries while maintaining structural integrity, thus improving their overall functionality in energy storage applications. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*

- **Rational Design of High-Voltage Lithium-Ion Batteries: From Multiscale Mechanisms to Scalable Manufacturing** — score: 0.800 This review examines the intricate relationship between fabrication techniques and the multiscale degradation mechanisms affecting high-voltage lithium-ion batteries (HVLIBs). It highlights the impact of advanced processing methods and electrode architecture on performance and stability, while discussing recent characterization and modeling innovations that link these factors. The authors aim to identify challenges and outline future directions for the rational design of HVLIBs in scalable manufacturing contexts. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Revealing the Mechanism for Boosted Oxygen Reduction Reaction of Distorted Iron Single Atom Electrocatalyst** — score: 0.800 This study introduces the Fe-N₃S/SNC electrocatalyst, characterized by a distorted structure that enhances the oxygen reduction reaction (ORR) performance. The mechanism involves stable electron filling in the eg orbital due to sulfur atoms acting as electron buffers, resulting in significantly improved catalytic activity compared to traditional Fe-N₄/NC and commercial Pt/C, as demonstrated through advanced spectroscopic and theoretical analyses. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Evolution of Charge and Orbital Ordering, and Cation Vacancy Ordering During Electrochemical Desodiation of Na_xNiO₂** — score: 0.800 The article investigates the changes in charge and orbital ordering, alongside cation vacancy ordering, that occur during the electrochemical desodiation process of Na_xNiO₂. Through this study, the authors reveal significant insights into the structural and electronic evolution of the material, which is crucial for its application in sodium-ion batteries. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Vibrationally-Resolved Electrochemical Impedance Spectroscopy** — score: 0.800 The article discusses a novel approach to electrochemical impedance spectroscopy that incorporates vibrational resolution, allowing for more detailed analysis of molecular interactions. This advancement can enhance understanding of electrochemical processes by providing insights into the vibrational states of the species involved, potentially leading to improvements in various applications in chemistry and materials science. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Effect of Nanoparticle Surface Modification on Lithium-Ion Transport in Composite Polymer Electrolytes** — score: 0.800 This article investigates the impact of surface modifications on nanoparticles incorporated into composite polymer electrolytes meant for solid-state lithium-metal batteries. The study highlights how variations in nanoparticle chemistry and functionalization influence both ionic conductivity and interfacial transport efficiency, which are critical for battery performance. Journal: *RSC - EES Batteries latest articles*
- **Bifunctional Sn/NaF Interlayer Enables Stable Solid State Sodium Batteries** — score: 0.800 The study introduces a bifunctional Sn/NaF interlayer at the interface of sodium solid-state batteries to enhance interfacial contact and suppress dendrite formation, achieving a fourfold increase in critical current density. The Na₃V₂(PO₄)₃@PEG/NZZP@NaF@Sn/Na batteries exhibit remarkable cycling stability, maintaining over 67% capacity retention after 10,000 cycles at elevated discharge rates, demonstrating the potential for improved solid-state battery performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Solid Electrolyte Interphase Formation Reactions on Lithium Metal with Ether-Based Electrolytes** — score: 0.800 The article investigates the formation reactions of the solid electrolyte interphase (SEI) on lithium metal when using ether-based electrolytes. It emphasizes the implications of these reactions for the performance and stability of lithium metal batteries, proposing insights into optimizing the SEI for improved battery efficiency. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Tailoring the Interfacial Behavior to Stabilize Iron Oxalate Anode for Boosting Ultrahigh Lithium Storage** — score: 0.800 This article discusses a novel method to enhance the performance of iron oxalate anodes for lithium-ion batteries by applying a carbon-rich coating from carboxymethyl

cellulose. This coating improves conductivity, stabilizes solid-electrolyte interphase (SEI) formation, and significantly boosts lithium storage capacity, achieving up to 1652 mAh g⁻¹ while maintaining 517 mAh g⁻¹ at high current densities, thereby offering a promising avenue for developing high-performance energy storage materials. Journal: *Wiley: Small Methods: Table of Contents*

- **Claw-Like Macromolecular Engineering for Stabilizing Zinc Anodes Under High-Current Operation** — score: 0.800 The research presents a novel electrolyte modification using tris(3,6-dioxahexyl)amine (TDA-1) with a claw-like structure to improve the stability of zinc anodes during high-current operations. By anchoring to the zinc surface and reconstructing the electric double layer, TDA-1 reduces desolvation barriers and promotes uniform zinc deposition. The implementation shows significant improvements in cycling stability for both symmetric and full batteries, demonstrating the effectiveness of macromolecular engineering in enhancing zinc anode performance. Journal: *Wiley: Small Methods: Table of Contents*
- **Multiscale Kinetics-Enhanced and Interphase-Stabilized Hierarchical Architecture Design Enables Fast-Charging and Longevous Sodium-Ion Batteries** — score: 0.800 This study introduces a novel bismuth@carbon composite anode for sodium-ion batteries, designed to enhance fast-charging and longevity. The unique microstructured architecture features densely packed nano-bismuth within nitrogen-doped carbon microrods, resulting in high electrochemical performance characterized by significant capacity retention over extensive cycling and rapid charge times. These findings suggest a promising strategy for next-generation battery technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Suppressed calendar aging losses via dynamic interfaces in metal anodes** — score: 0.700 This study explores a method to reduce calendar aging in metal anodes, particularly magnesium, by creating a dynamic, reversible interface. This innovative approach can enhance the long-term viability of reactive metal anodes, addressing a significant limitation in their practical application and boosting their energy storage potential. Journal: *Joule*
- **Anion-reduction catalytic centres regulate interfacial solvation structures for fast-charging Si anodes** — score: 0.700 The research by Qiao et al. highlights the role of S-vacancy-driven anion-reduction catalysis in regulating interfacial solvation structures at silicon anodes, resulting in the formation of a LiF-rich solid electrolyte interphase. This innovation allows for both ultrafast charging capabilities and an extended cycle life for batteries. Journal: *Nature Energy*
- **Decoupling and Cooperating Interfacial and Bulk Transport via a Bio-inspired Molecular Modulator for High-Performance All-Solid-State Li Batteries** — score: 0.700 The article explores a bio-inspired molecular modulator designed to enhance transport mechanisms in all-solid-state lithium batteries by independently improving interfacial and bulk transport. This innovation aims to achieve high-performance battery operation, addressing current limitations in energy storage technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] Polymer-Modulated Solvation Chemistry via Compatibilizing-Solvent Plasticization for Stable High-Energy Lithium Metal Batteries** — score: 0.600 The article discusses a novel approach to enhance the stability of high-energy lithium metal batteries by utilizing compatibilizing solvents to modulate solvation chemistry through polymer plasticization. This technique aims to improve battery performance and longevity, addressing critical challenges in lithium metal battery technology. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Local Coordination Environment Engineering of Na₃ Sites in Na₄Mn_{1.5}Fe_{1.5}(PO₄)₂P₂O₇ Cathode** — score: 0.600 The article discusses the engineering of the local coordination environment of sodium (Na₃) sites in the cathode material Na₄Mn_{1.5}Fe_{1.5}(PO₄)₂P₂O₇. This research aims to enhance the electrochemical performance of the cathode by optimizing the structural characteristics, potentially leading to improvements in sodium-ion battery technology. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Nature Seed Pods-Inspired Microstructure Toward Thermal-Mechanical-Electrochemical Robust Separators for High-Safety Lithium Metal Batteries** — score: 0.600 The article

presents a bio-inspired separator for lithium metal batteries, mimicking the protective structure of seed pods to enhance safety and longevity. The aluminum nitride and aramid nanofiber-coated polypropylene separator effectively blocks lithium dendrite formation, improves thermal management, and regulates lithium deposition, resulting in pouch cells that maintain 88.26% capacity after 500 cycles. This design offers a significant advancement in developing safe and efficient high-energy lithium metal batteries. Journal: *Wiley: Small Methods: Table of Contents*

- **Engineering Interfacial Built-In Electric Fields for Accelerated and Stable Sodium-Ion Storage** — score: 0.600 The article discusses a novel approach to enhance the performance of sodium-ion batteries by engineering interfacial built-in electric fields (BEF). This method aims to accelerate sodium-ion storage while improving stability, addressing challenges related to the controllability of BEF intensity to optimize electrochemical efficiency. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Probing Into the NiO Segregation Mechanism for Optimized Synthesis of High-Capacity Sodium-Ion Layered Cathodes** — score: 0.600 This article examines the segregation mechanism of NiO impurities in high-Ni sodium-ion layered cathodes during high-temperature synthesis, which hampers capacity. The authors propose a novel low-temperature annealing process with excess sodium to reintegrate NiO, resulting in a phase-pure cathode that achieves a reversible capacity of 156 mAh/g and demonstrates strong cycling stability, indicating significant potential for high-energy sodium-ion batteries. Journal: *Wiley: Small Methods: Table of Contents*
- **Atomistic-to-Device Understanding of Oxygen Reduction Using Redox-Active Lanthanide-Coupled Carbon Nitride Nanoparticles for Bio-Electrochemical Power Generation in Microbial Fuel Cells** — score: 0.600 The study presents a novel approach using lanthanide-coupled graphitic carbon nitride nanoparticles to enhance oxygen reduction reactions in microbial fuel cells. By incorporating rare-earth ions ($\text{Ce}^{3+}/\text{Gd}^{3+}$), it achieves defect-rich, high-performance catalysts that significantly improve power output and chemical oxygen demand removal, outperforming conventional platinum catalysts while maintaining stability and efficiency for sustainable energy applications. Journal: *Wiley: Small Methods: Table of Contents*
- **[ASAP] Ordered–Disordered Ionic Cocrystalline Solid-State Electrolytes for Rapid Ion Migration in Sodium Metal Batteries** — score: 0.500 The article discusses the development of ordered-disordered ionic cocrystalline solid-state electrolytes that enhance rapid ion migration in sodium metal batteries. This innovation aims to address the challenges of ion transport efficiency, potentially leading to improved performance and safety in energy storage applications. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Asymmetric Glycol-Ether Molecule Design Enables Temperature-Adaptive Anion Coordination for Durable High-Temperature Sodium Batteries** — score: 0.400 The article discusses the innovative design of asymmetric glycol-ether molecules that facilitate temperature-adaptive coordination with anions, contributing to the enhanced performance and durability of high-temperature sodium batteries. This advancement aims to improve the efficiency and stability of energy storage solutions in elevated thermal conditions. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Lignin-Functionalized Supramolecular Binder Enables Aggressive Cathode Chemistries in Advanced Li-Ion Batteries** — score: 0.400 The article discusses the development of a lignin-functionalized supramolecular binder that enhances the performance of advanced lithium-ion batteries. By improving the stability and effectiveness of aggressive cathode materials, this innovative binder holds potential for increasing the efficiency and longevity of battery technologies. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Bismuth-Based Heterojunctions with a Built-In Electric Field Boosting High-Efficiency Polysulfide Conversion in Lithium–Sulfur Batteries** — score: 0.400 This article presents the development of bismuth-based heterojunctions that incorporate a built-in electric field, which significantly enhances the conversion efficiency of polysulfides in lithium-sulfur batteries. The innovative material design aims to address performance limitations in energy storage applications.

Journal: *Energy & Fuels: Latest Articles (ACS Publications)*

- **[ASAP] Synergistic Lattice and Interface Engineering of Ti-Doped LiFe_{0.5}Mn_{0.5}PO₄/C Cathodes from Spent Lithium-Ion Batteries** — score: 0.400 The article discusses the innovative approach of enhancing the performance of Ti-doped LiFe_{0.5}Mn_{0.5}PO₄/C cathodes, derived from recycled lithium-ion batteries, by employing synergistic lattice and interface engineering techniques. This research aims to improve electrochemical properties and efficiency, contributing to the sustainability of battery materials. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Dual Soft-Gel Electrodes Enabling Near-Practical and Deployable Aqueous Batteries** — score: 0.400 This article presents a novel dual soft-gel electrode aqueous battery that utilizes a water-competition effect to stabilize redox-active species, resulting in improved electrochemical performance. The research demonstrates that cation hydration affects the stability and water content of the soft gels, and the concept is scalable to a cylindrical battery design, showing promise for practical deployment in energy storage applications. Journal: *Wiley: Small Methods: Table of Contents*
- **[ASAP] Isomer Engineering toward High-Voltage and Air-Stable Organic Cathodes for Lithium-Ion Batteries** — score: 0.300 The article discusses advancements in isomer engineering aimed at developing high-voltage, air-stable organic cathodes for lithium-ion batteries. It highlights the potential of these engineered materials to enhance battery performance and stability, offering a promising avenue for improving energy storage technologies. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **Thermodynamically Guided Three-Stage Separation for Deep Aluminum Removal from Delithiated LiFePO₄ Slag Toward Battery-Grade FePO₄ Regeneration** — score: 0.300 This study introduces a three-stage method for efficiently removing aluminum from delithiated LiFePO₄ slag to regenerate high-purity FePO₄ suitable for battery applications. By leveraging selective leaching, controlled precipitation, and recrystallization, the process achieves a significant aluminum reduction, obtaining FePO₄ with less than 300 ppm aluminum, thereby advancing closed-loop recycling in battery materials. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Thermal energy storage materials for suppressing thermal runaway propagation in lithium-ion batteries: a review** — score: 0.300 The article reviews recent developments in thermal energy storage materials aimed at preventing thermal runaway in lithium-ion batteries. It discusses various enhancement strategies and the potential for integrating these materials with active cooling systems to improve battery safety and performance. Journal: *RSC - EES Batteries latest articles*
- **Coordinated Polymer Electrolytes Achieve 5.1 V Stability with Rapid Ion Transport for High Performance Lithium Metal Battery** — score: 0.300 The article discusses the development of coordinated polymer electrolytes that enhance the stability of lithium metal batteries to 5.1 V while also improving the rate of ion transport. This advancement represents a significant step towards achieving high-performance energy storage solutions, potentially increasing the efficiency and longevity of lithium metal batteries. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] Overlooked Decomposition of Organic Cosolvents Undermines Stability of Aqueous Hybrid Electrolytes** — score: 0.300 The article highlights that the degradation of organic cosolvents in aqueous hybrid electrolytes is a significant yet often overlooked factor that adversely affects their stability. This decomposition process could lead to reduced performance and reliability in energy storage applications. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **[ASAP] Organic-Solvent-Free Weakly Solvating Electrolytes Enable Durable Ah-Scale Zn-Iodine Batteries under Diverse Harsh Operating Conditions** — score: 0.300 The article discusses the development of organic-solvent-free weakly solvating electrolytes that enhance the performance and durability of zinc-iodine batteries, particularly under challenging operating conditions. This advancement opens up new possibilities for the use of such batteries in various applications, contributing to more sustainable energy storage solutions. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*

- **Entropy-driven electronic reconfiguration in high-entropy sulfides enables rapid charge-transfer dynamics for next-gen wearable energy devices** — score: 0.200 The study explores high-entropy sulfides that leverage entropy-driven electronic reconfiguration to enhance charge-transfer dynamics. This advancement is positioned as a significant step toward the development of next-generation energy storage solutions suitable for wearable devices. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Recent Progress on Natural Biopolymer-Based Electrolytes for Advanced Batteries** — score: 0.200 This article reviews the advancements in natural biopolymer-based electrolytes for next-generation batteries, highlighting their potential benefits in improving battery performance and sustainability. The authors discuss the latest research findings and applications, emphasizing the role of these eco-friendly materials in enhancing energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] Hydrophobic–Hydrophilic Balance in Copolymer Binder Enables High-Loading, High-Rate, and Stable Flexible Aqueous Zinc-Ion Batteries** — score: 0.200 The article explores the role of a copolymer binder that balances hydrophobic and hydrophilic properties, enhancing the performance of flexible aqueous zinc-ion batteries. It highlights how this balance enables high loading capacity, increased charge rates, and stable operation, paving the way for more efficient energy storage solutions. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Ultrahigh Te-Content Low-Pressure All-Solid-State Li–Te Batteries** — score: 0.200 The article discusses the development of ultrahigh tellurium-content low-pressure all-solid-state lithium-tellurium batteries, emphasizing their potential to enhance performance and safety compared to conventional battery systems. The findings suggest advancements in energy storage technology through innovative material combinations. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Three scientists pushing chemistry in new directions** — score: 0.200 The article highlights three prominent scientists who are advancing the field of chemistry through groundbreaking research, focusing on enhancements in battery technology and accelerating reaction processes. Their contributions are set to significantly influence both academic and practical applications in the industry. Journal: *Nature*
- **CO₂-enabled recovery** — score: 0.200 The article discusses a novel CO₂-assisted recovery protocol that effectively removes cation contaminants from platinum-alloy catalysts in proton exchange membrane fuel cells. This breakthrough could significantly enhance the performance and commercial viability of these catalysts, addressing a longstanding challenge in the field. Journal: *Nature Energy*
- **Efficient Nitrate Conversion to NH₃ on Core/Shell-Like Structured Cu–Co Heterogeneous Catalyst With Surface-Enriched Atomically-Dispersed Co Dopant** — score: 0.200 The article presents a newly synthesized core/shell Cu–Co heterogeneous catalyst designed for efficient electrocatalytic nitrate reduction to ammonia (NH₃). Utilizing CoSn(OH)₆ as the core and atomically dispersed Co-doped CuSn(OH)₆ as the shell, the catalyst showcases excellent performance with a Faradaic efficiency of 79.19% for NH₃ production and is suitable for a high-efficiency Zn–NO₃–battery, effectively addressing environmental challenges through sustainable resource conversion. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Bipolar NASICON Electrodes for Symmetric Sodium-Ion Batteries: An Outlook** — score: 0.200 The article discusses the potential of bipolar NASICON (Na Super Ionic Conductor) electrodes for enhancing the performance of symmetric sodium-ion batteries. It highlights advances in materials that could improve energy efficiency and stability in battery applications, outlining directions for future research in this area. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **[ASAP] Chelation-Driven Phosphorus Doping: Upcycling Fungal Industrial Waste into Hard Carbon Anodes** — score: 0.200 The article discusses a method for converting industrial fungal waste into hard carbon anodes through a chelation-driven phosphorus doping process. This approach not only addresses waste management but also enhances the performance of carbon anodes in energy storage applications. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*

- **[ASAP] Hydrophobic Solvation-Driven Stabilization of the Fluorenone Radical for the Anolyte of All-Organic Flow Batteries under Benign pH Conditions** — score: 0.200 The article discusses the hydrophobic solvation effects that contribute to the stabilization of the fluorenone radical, which is utilized as the anolyte in all-organic flow batteries. This research highlights the benefits of maintaining benign pH conditions for optimizing battery performance, underscoring the importance of solvation dynamics in electrochemical systems. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **A thin and robust polymer electrolyte membrane with dynamic metal-ion crosslinking** — score: 0.200 The article discusses the development of a polymer electrolyte membrane made from polybenzimidazole that utilizes dynamic copper-ion crosslinking, resulting in a thin and mechanically strong material. Fuel cells using this membrane demonstrate impressive peak power density and stability under high-temperature conditions. Journal: *Nature Energy*
- **A New Paradigm for Flowless Zn–Br₂ Aqueous Cathodes: Fast Kinetics and Stability via Entrapped Phosphonium Bromides** — score: 0.200 The article presents a novel approach to zinc-bromine (Zn–Br₂) aqueous cathodes, utilizing entrapped phosphonium bromides to enhance the kinetics and stability of the batteries without the need for flow systems. This innovation addresses efficiency challenges in energy storage, potentially improving the performance and longevity of Zn–Br₂ batteries. Journal: *ScienceDirect Publication: Energy Storage Materials*